ModelIt!

Student Activity Pack

Carbon Capture at Scale: Can We Vacuum the Atmosphere?

Modeling the Engineering and Economics of Removing CO2 from the Sky

Grade 10L3 | NGSS HS-ESS2-6, HS-PS1-4 | Scale, Proportion, and Quantity

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What do you think would happen if we could suck CO2 directly out of the atmosphere?

2 Why might removing CO2 from the air be harder than preventing it from being emitted in the first place?

3 Draw what you think a carbon capture system would look like and how it would work.

4 If carbon capture technology exists, why have we not solved climate change already?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

CO2 Concentration | Capture Rate | Sorbent Efficiency | Energy Input | Storage Capacity | Reaction Temperature | Regeneration Cost | Transport Distance | Geological Sequestration | Carbon Credit Value

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Current Technology

2024-era capture technology at 420 ppm with fossil energy mix

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Clean Energy Powered

All operations powered by renewable energy with zero emissions

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Scaled to Climate Impact

Operations scaled to 1 billion tons per year of capture

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Sorbent Lifetime — The number of capture-regeneration cycles a sorbent material can undergo before degrading to the point of replacement — longer lifetimes reduce material costs and waste

Water Consumption — The volume of water used in capture processes and cooling systems — a significant concern in arid regions where water scarcity may limit facility siting

CO2 Utilization Revenue — Income from selling captured CO2 for industrial use — concrete curing, synthetic fuels, carbonated beverages — which can offset capture costs but does not permanently remove CO2 from the carbon cycle

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Can technology remove enough CO_2 from the atmosphere to actually reverse climate change — or is it just an expensive distraction? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate the critical importance of the energy source in determining whether carbon capture is a net climate benefit?**

2. **What does the cost per ton comparison between direct air capture and other CO_2 removal methods reveal about the role of each in climate strategy?**

3. **Why might carbon capture be necessary even if it is more expensive than simply reducing emissions?**

4. **How would your model change if carbon credit prices rose to $500 per ton globally?**

5. **What are the ethical implications of relying on carbon capture technology to address climate change versus reducing emissions at the source?**

8 STEM Challenge: Design a Carbon Capture Business Plan

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YOUR ENGINEERING MISSION

Design a commercially viable direct air capture facility that maximizes net CO_2 removal while achieving economic sustainability through carbon credit sales.

THE SCENARIO

A climate technology venture fund is evaluating proposals for next-generation direct air capture facilities. Your team must design a facility that specifies the capture technology, energy source, geological storage location, and business model — demonstrating that the operation can remove CO_2 profitably at a carbon credit price of $250 per ton within 10 years.

Engineering Design Thinking

1. WHAT CAPTURE TECHNOLOGY DOES YOUR FACILITY USE AND WHAT IS ITS SORBENT EFFICIENCY?
2. HOW WILL YOU SUPPLY THE THERMAL ENERGY NEEDED FOR SORBENT REGENERATION WITHOUT FOSSIL FUELS?
3. WHAT GEOLOGICAL FORMATION WILL YOU USE FOR STORAGE AND HOW WILL YOU ENSURE PERMANENCE?
4. WHAT IS YOUR PROJECTED COST PER TON AT YEAR 1 VERSUS YEAR 10, AND WHAT DRIVES THE COST REDUCTION?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Carbon Capture Engineers design and optimize direct air capture systems, sorbent materials, and geological storage operations. They work at Climeworks, Carbon Engineering, Heirloom Carbon, and major energy companies, earning $80,000-$150,000/year. Carbon Market Analysts who evaluate capture projects for credit certification earn $70,000-$130,000/year.